

CS 4803/8803: Sociotechnical Security

Course Information

Term: Fall 2026

Credit Hours: 3

Instructor: Veronica Rivera (varivera@gatech.edu)

Office hours: Wednesdays 4:00 - 5:00 PM on the second floor of Coda, or by appointment.

Course Description

This course will focus on how human factors, such as human behavior, goals, and needs, both shape and are shaped by, the security and privacy of digital systems. For instance, how human behavior can lead to insecure systems and how digital systems can threaten users' safety. We will explore various topics in usable security and privacy and digital safety such as authentication, mental models of encryption, scams, online hate and harassment, and technology-facilitated interpersonal abuse. We will learn and apply human-centered and empirical research methods in the context of these topics, and discuss best practices of human-centered and empirical research in computer security.

Course Learning Outcomes

Upon successful completion of the course, you should be able to:

1. Speak knowledgeably about topics at the intersection of human behavior, security & privacy, and technology design.
2. Identify potential failures of digital systems through a human-centered security lens.
3. Evaluate the impact of digital systems on individual and societal level security, privacy, and safety
4. Apply human-centered and empirical research methods to design for and evaluate security, privacy, and safety, in digital systems.

Required Materials

No textbooks are required. Instructor will provide any necessary reading material on course website.

The following will be used in the course:

- Course website:
 - Schedule, assignment instructions, reading assignments. Check this often.
- Ed:
 - For announcements and questions related to the course
- Gradescope:
 - For submitting all course assignments

Prerequisites

Undergraduate students must have taken CS 3237 (Human Dimension of Cybersecurity: People, Organizations, Societies).

There are no prerequisites for graduate students. Having taken a course in Human-Computer Interaction (HCI) is helpful, but not strictly required.

Coursework and Grades

The grade cutoffs will be approximately: A > 90%; B > 80%; C > 70%; D > 60%.

The instructor reserves the right to adjust these upwards or downwards depending on how hard the assignments and course project end up being.

Graded Components

There are three primary assignments in this course. Detailed instructions for all deliverables will be linked on the course website.

Attendance and participation (15%):

We will be discussing the assigned research papers each class period. Therefore, students are expected to attend each class session and contribute meaningfully to the discussion. Participation will be tracked by instructor. Students will be allowed to miss up to 2 class sessions without impacting their grade. If other class sessions must be missed, please contact the instructor ahead of the class session to make alternative arrangements.

Reading response (10%):

For each class period, students will submit a short reading response to prepare for that day's lecture/discussion. Reading responses will be due at **2:00 PM** on the day of class. Because the reading response is meant to help students prepare for that day's class, **no late reading responses will be accepted**. Responses are graded for depth and thoughtfulness, and students offering their own independent judgment over precise summaries. The two lowest reading response grades will be dropped.

Leading discussion (25%):

Students will lead the discussion on the assigned readings twice throughout the semester. For this, they will be expected to prepare a 20 minute presentation and 5 discussion questions. They will sign up for their preferred presentation date in week 2. **Students will be expected to share the slides and discussion questions with Veronica at least 3 days in advance and iterate with her on developing the final slides before the presentation date.** Students are exempt from submitting a reading response on the days they are leading discussion. Students will be evaluated on the fair representation of the points in the work, presentation style, and ability to facilitate dialogue with their peers.

Course project (50%):

Students will complete a research project in an area of sociotechnical security and submit intermediate deliverables.

- Project proposal. **Due 9/9 at 2:00 PM.**
 - This document will commit your group members and the general project direction you are going in. You will sign up for a meeting with the instructor after the project proposal is submitted.
- Project progress report. **Due 10/14 at 2:00 PM.**
 - This 2-page document describes your current status on the project. You will sign up for a meeting with the instructor after the progress report is submitted
- Final Paper. **Due 12/7 at 11:59 PM any where on earth.**
 - The final report should be 5-6 pages and include: a clear problem statement with motivation for studying the problem, a related work survey, a description of your study approach, and your results. If you built a system, such as for measuring abuse in a sociotechnical system, you should include the process by which your system was built and tested when discussing your approach. Final reports should be submitted using the USENIX LaTeX template
- Final Presentation. **12/7 in class.**
 - Each project group will present for 30 minutes (20 minute presentation + 10 minute Q&A). Instructions for the final presentation will be provided closer to the date.

Generative AI Policy

Students can use LLMs for checking spelling and grammar and as they would a search engine (e.g., finding related work), but they cannot be used to produce any deliverable for this course. That includes reading papers, writing reading responses, making presentations, analyzing data, and building systems. This is a course about identifying new ways systems fail and cause harm and how to solve these problems. LLMs won't offer any new insight here; they are trained on existing work, and this is a problem existing work still hasn't solved. If you are unsure of whether or not you can use an LLM for a particular task in this course, best to just ask the instructor. Failure to follow these policies will be considered a violation of academic integrity.

Extensions and Late/Missed Assignments Policy

You may miss up to 2 classes without telling me in advance. If you have additional conflicts please contact the instructor. Missing additional classes without talking to the instructor in advance will result in a reduction of your grade.

Deadlines for all other assignments are firm. Late reading responses will not be accepted. For project deliverables, your grade will be reduced **10%** for every 24 hours beyond the deadline period.

If you have a pressing emergency, like a personal conflict, illness, or other circumstance, please reach out to the instructor to discuss.

I also recognize that many of you are taking other demanding classes and have responsibilities outside the school that might cause you to fall behind. If you find yourself in this position, or struggling to keep up, please come talk to me. We can work together to find a suitable arrangement for you to learn and be successful in this class.

Course Policies

Attendance

Attendance in class is mandatory and students will be expected to show up prepared and ready to engage. Lectures will not be recorded. Lecture slides will be released following each class period, but active participation in class is required for success in the course.

However, **please do not attend the class in person if you are ill**. If you are ill and need to stay home, please email the instructor as soon as possible for alternative arrangements. The same policy applies if you need to be absent for emergencies or extenuating circumstances – please email the instructor **before the class you will miss** explaining your circumstance to discuss alternatives. Students are allowed 2 unexcused absences without impact to their grade.

Academic Integrity

Georgia Tech aims to cultivate a community based on trust, academic integrity, and honor. Students are expected to act according to the highest ethical standards. Review Georgia Tech's [Honor Code](#) and the student [Code of Conduct](#).

Any student suspected of cheating or plagiarism on a quiz, exam, or assignment will be reported to the Office of Student Integrity, who will investigate the incident and identify the appropriate penalty for violations.

Student-Faculty Expectations

At Georgia Tech, we believe that it is important to strive for an atmosphere of mutual respect, acknowledgement, and responsibility between faculty members and the student body. The [Student-Faculty Expectations](#) articulates some basic expectations that you can have of me and that I have of you. In the end, simple respect for knowledge, hard work, and cordial interactions will help build the

environment we seek. Therefore, I encourage you to remain committed to the ideals of Georgia Tech while in this class.

Communication with Professor

You can contact me via email for anything related to the course. Please include CS 8803 and the topic of your message in the subject line. I receive many emails each day. **If you do not include CS 8803 in the subject, there is a high chance I will not read your email.** You can expect a response within 3 business days. Do not expect an immediate response to your email after business hours, during the weekend, university holidays, or while I am traveling. **Before emailing, make sure that your question has not already been addressed by the syllabus or announcements.**

Accommodations for Students with Disabilities

If you are a student with learning needs that require special accommodation, contact the [Office of Disability Services](#) (404-894-2563) as soon as possible to make an appointment to discuss your special needs and to obtain an accommodations letter. Please also e-mail me as soon as possible in order to set up a time to discuss your learning needs.

Collaboration

You can discuss and work on all assignments with others unless explicitly noted in the assignment instructions. If you choose to discuss and work with others, (1) any work you turn in must be written in your own hand, and (2) your submission must acknowledge your collaborators and how you received help from them. For example “I worked with Alice and Bob on writing my interview protocol for this assignment”.

Acknowledgements

This class is inspired by a number of other classes at other institutions including Georgetown COSC 3720 (Sociotechnical Security), NYU CS 3943 (Trust and Safety), and UC San Diego CSE 291D (Sociotechnical Cybersecurity)